

CALCULUS III – PRACTICE TEST 1

Time: 90 minutes

Total: 10 points

Full name: _____ Student ID: _____

No notes, textbook, or electronic devices are permitted.

Question 1 (2 points).

- a) Find the limit, if it exists, or show that it does not exist:

$$\lim_{(x,y) \rightarrow (1,-2)} \frac{x^2y + y^2}{x^2 + y^2}.$$

- b) Find the limit, if it exists, or show that it does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^4 + y^2}.$$

- c) Determine the value of a for which the function is continuous at $(0, 0)$:

$$f(x, y) = \begin{cases} \frac{x^2y^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ a, & (x, y) = (0, 0). \end{cases}$$

Question 2 (3 points).

- a) Let $z = x^2 \cos y + ye^x$, where $x = s^2 + t$ and $y = st$. Use the Chain Rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $(s, t) = (1, 0)$.
- b) If $x^2 + y^2 + z^2 + xyz = 7$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
- c) Find the directional derivative of $f(x, y) = x^2e^y$ at $(1, 0)$ in the direction from $(1, 0)$ toward $(4, 4)$.

Question 3 (3 points).

Find the absolute maximum and absolute minimum values of

$$f(x, y) = x^2 + y^2 - 2x + 4y$$

on the closed disk $x^2 + y^2 \leq 16$.

Question 4 (2 points).

Find the volume of the solid under the surface $z = 2 + x + y^2$ and above the region enclosed by $x = y^2$ and $x = 2y$.

CALCULUS III – PRACTICE TEST 2

Time: 90 minutes

Total: 10 points

Full name: _____ Student ID: _____

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Question 1 (2 points).

a) Evaluate

$$\lim_{(x,y) \rightarrow (2,1)} \frac{xy^2 - 2}{x + y - 2}.$$

b) Determine whether the following limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - 3xy^2}{x^2 + y^2}.$$

c) Find a so that f is continuous at $(0, 0)$:

$$f(x, y) = \begin{cases} \frac{\sin(x^2 + y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ a, & (x, y) = (0, 0). \end{cases}$$

Question 2 (3 points).

a) Let $w = \ln(x^2 + y^2)$, where $x = u \cos v$ and $y = u \sin v$. Find $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ at $(u, v) = (2, \pi/4)$.

b) If $e^{xz} + y^2z = x + 1$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

c) Find the directional derivative of $f(x, y) = \ln(x^2 + y^2)$ at $(1, 1)$ in the direction of the vector $\mathbf{v} = (-3, 4)$.

Question 3 (3 points).

Find the absolute maximum and absolute minimum values of

$$f(x, y) = 3x - 4y$$

on the disk $x^2 + y^2 \leq 25$.

Question 4 (2 points).

Find the volume under $z = xy + 4$ and above the region bounded by $y = x^2$ and $y = 2x$.

CALCULUS III – PRACTICE TEST 3

Time: 90 minutes

Total: 10 points

Full name: _____ Student ID: _____

No notes, textbook, or electronic devices are permitted.

Question 1 (2 points).

a) Evaluate

$$\lim_{(x,y) \rightarrow (-1,2)} \frac{x^2 + xy + y^2}{1 + x^2 + y^2}.$$

b) Find the limit, if it exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}.$$

c) Determine a so that f is continuous at the origin:

$$f(x, y) = \begin{cases} \frac{x^3 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ a, & (x, y) = (0, 0). \end{cases}$$

Question 2 (3 points).

a) Let $z = e^{xy} + x \sin y$, where $x = r + t^2$ and $y = rt$. Find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial t}$ at $(r, t) = (0, 1)$.

b) If $x \cos z + y \sin z = 2$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

c) Find the directional derivative of $f(x, y) = \sqrt{x^2 + 2y^2}$ at $(2, 1)$ in the direction from $(2, 1)$ to $(-1, 5)$.

Question 3 (3 points).

Find the absolute extrema of

$$f(x, y) = x^2 + 2y^2 - 4x$$

on the closed disk $x^2 + y^2 \leq 9$.

Question 4 (2 points).

Find the volume under the surface $z = 1 + x^2 + y$ and above the region enclosed by $y = x^2$ and $y = 2 - x^2$.

CALCULUS III – PRACTICE TEST 4

Time: 90 minutes

Total: 10 points

Full name: _____ Student ID: _____

No notes, textbook, or electronic devices are permitted.

Question 1 (2 points).

a) Evaluate

$$\lim_{(x,y) \rightarrow (0,2)} \frac{e^{xy} - 1}{x}.$$

b) Determine whether the following limit exists:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}.$$

c) Find the value of a that makes f continuous at $(0,0)$:

$$f(x, y) = \begin{cases} \frac{1 - \cos(x^2 + y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ a, & (x, y) = (0, 0). \end{cases}$$

Question 2 (3 points).

a) Let $w = x \ln y + y \cos x$, where $x = pq$ and $y = p + q$. Use the Chain Rule to find $\frac{\partial w}{\partial p}$ and $\frac{\partial w}{\partial q}$ at $(p, q) = (1, 1)$.

b) If $x^2 z + y z^2 = 6$, find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

c) Find the directional derivative of $f(x, y) = \sin(xy)$ at $(\pi/2, 1)$ in the direction of $\mathbf{v} = (5, 12)$.

Question 3 (3 points).

Find the absolute maximum and absolute minimum values of

$$f(x, y) = xy$$

on the disk $x^2 + y^2 \leq 8$.

Question 4 (2 points).

Find the volume under $z = 3 - x + y^2$ and above the region enclosed by $x = y^2 - 1$ and $x = 3 - y^2$.